

Reduction of *Fusarium* mycotoxins in maize by ozonation: A sustainable approach to feed storage and evaluation of maize quality parameters

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Highlights

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Ozonation allowed the reduction of the *Fusarium*-generated mycotoxins in maize grain.

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During the ozonation procedure the enzymatic activity of the maize grain decreased.

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The oxidation process resulted in the modification of the fatty acid profile.

Abstract

Maize (*Zea mays* L.) is a crop of global importance as food and feed. Among the most hazardous contaminants responsible for its quality degradation are species of the fungi genera *Fusarium*, *Aspergillus*, and *Penicillium*, which synthesize toxic secondary metabolites such as deoxynivalenol, fumonisins, HT2/T2 toxins and zearalenone.

In this study, we investigate the efficiency of the detoxification of mycotoxin-contaminated maize grains by ozonation and its impact on the quality parameters of the maize in short and long storage, depending on the concentration of ozone and the exposition time. Maize grains were artificially inoculated with three species of *Fusarium* (*F. avenaceum*, *F. graminearum*, and *F. proliferatum*) to generate a spectrum of mycotoxins. In particular, *F. proliferatum* exhibited the highest mycotoxin production capacity, while *F. avenaceum* and *F. graminearum* produced fewer metabolites. Subsequent exposure to ozone at concentrations of 150–300 ppm for 15–30 min resulted in complete degradation of all detected mycotoxins and no intermediate degradation products were observed. Although ozonation was effective in removing toxins, its effect on maize quality parameters

was evident. The level of fatty acid oxidation marker (MDA) increased with increasing ozone dose, and the effect intensified with storage duration. This indicated a reduction in lipids as a result of exposure to ozone. The profile of fatty acids was also found to be affected by ozone; the content of 9-Octadecaenoic acid showed a negative correlation with storage time, while 9,12-Octadecadienoic acid exhibited a positive correlation with storage time. Furthermore, exposure to ozone caused the Lipoxygenase and Proteolytic activity to decrease. This effect, however, was found to be reversible; after long storage duration, both parameters regained their initial values. The results obtained underscored the need for careful optimization of treatment conditions.

Introduction

Maize (*Zea mays* L.) is one of the most widely cultivated cereal crops worldwide and serves as a fundamental raw material in both human nutrition and animal feed (Carvajal-Moreno, 2022; Nyandi and Pepó, 2024). It is consumed in various forms, from fresh grains to processed products such as silage, flour, and starch. However, as with other cereal grains, maize is highly susceptible to contamination by a variety of pests and pathogens, particularly fungi, during cultivation, harvest, and storage (Kortei et al., 2022; Odjo et al., 2022).

Among the most concerning fungal contaminants are species belonging to three main genera: *Aspergillus*, *Fusarium*, and *Penicillium*, which thrive in the nutrient rich environment provided by maize (Afsah-Hejri et al., 2020; Dipendra et al., 2019; Mohamed et al., 2023). These fungi are capable of producing a wide array of secondary metabolites known as mycotoxins, including deoxynivalenol (DON), fumonisins, HT-2 and T-2 toxins and zearalenone (Binder et al., 2007; Czembor et al., 2024; Ekwomadu et al., 2021; Glenn, 2007; Placinta et al., 1999). These compounds pose significant risks to human and animal health, as they exhibit acute and chronic intoxication, due to their carcinogenic, immunosuppressive, and endocrine disrupting properties. In addition, intoxication with various mycotoxins results in reproductive disorders. (Carvajal-Moreno, 2022; Greco et al., 2014; Lv et al., 2025; Odjo et al., 2022; Tassis et al., 2022).

According to estimates from the American Society of Phytopathology, approximately 80% of economically important plant species are affected by various strains of *Fusarium* (Bahadur, 2022; Esamaldeen et al., 2023). In the case of crops, even a quarter of their production in the world can be infected by fungi and contaminated by their metabolites each year (Atnafu et al., 2024; Mohamed et al., 2023). The presence of fungi on maize grain used for animal feed can be transferred to animal originated products such as meat, milk, or eggs that humans consume. For example, poultry, even if they survive exposure to mycotoxins in food, are susceptible to the accumulation of these toxins and

may consequently produce eggs or meat with elevated levels of mycotoxins (Gherbawy et al., 2024; Pascale et al., 1999). Furthermore, milk and dairy products from cattle fed plants contaminated with mycotoxin can contain high levels of these toxins. Therefore, feed contaminated with mycotoxins are finally harmful not only to animals but also to humans (Becker-Algeri et al., 2016; Greco et al., 2014; Pires et al., 2025).

To mitigate fungal contamination and mycotoxin production, various strategies have been used, with chemical fungicides being the most common (Habschied et al., 2021; Mielniczuk and Skwaryło-Bednarz, 2020). However, the widespread use of fungicides raises serious concerns due to their potential toxicity to non-target organisms, including humans. Prolonged or excessive exposure to certain fungicidal compounds has been associated with adverse health effects such as skin and respiratory irritation, organ damage, hormonal imbalances, and an increased risk of cancer (Odjo et al., 2022; Tao et al., 2024). Furthermore, the development of fungal resistance and the environmental persistence of these chemicals further complicate their long-term efficacy and safety.

In light of these challenges, there is growing interest in alternative and sustainable methods to reduce fungal contamination and mycotoxin levels in maize. One of such promising approaches is ozonation, a process that uses ozone (O_3), a powerful oxidizing agent, to degrade mycotoxins and inhibit fungal growth (Mielniczuk and Skwaryło-Bednarz, 2020; Purar et al., 2022; Udomkun et al., 2017). This method offers several advantages, including minimal chemical residue, broad-spectrum antimicrobial activity, and potential applicability during post-harvest storage (Freitas-Silva and Venâncio, 2010; Odjo et al., 2022). Trombete et al. (2017) reported the effect of mycotoxin-contaminated wheat ozonation with a concentration of 60 ppm O_3 at an exposure time of 300 min. Reihnolds (2016) performed ozonation of wheat with the ozone concentrations of 20 ppm O_3 for 40 and 130 min. Antos et al. (2024) used ozone with a concentration in the range of 500-3000 ppm for 60 min, to remove a range of mycotoxins generated by *Aspergillus* strains in maize grains. The most complex research on mycotoxin-contaminated maize ozonation to date was performed by Purar et al. (2022), who performed ozonation with ozone concentration in the range of 40-85 ppm for 180 min. In that study, aside of mycotoxin degradation, some basic research on ozone-induced oxidation of saturated and unsaturated fatty acids was conducted.

In the present study, we determined the effectiveness of ozonation for a relatively short exposure time by applying ozone concentrations in the range of 150 to 300 ppm for 15 to 30 min, which was much shorter than the exposure time used in other studies. We aimed to determine the impact of ozone dosing on various parameters of the maize grain, including the composition of fatty acids, the enzyme activity, and the level of

malondialdehyde (MDA), which is a biological marker of oxidative stress in maize grains. The latter two analyses have not been done so far. To examine the effect of time lapse on changes in quality parameters, we also determined the parameters of maize grains directly after ozone treatment and after long storage, i.e. after 1 and 2 months. Such an effect has also not been examined until now. The research was carried out for maize grains colonized with three different *Fusarium* strains, which are often responsible for maize intoxication (Purar et al., 2022; Velluti et al., 2000). This allowed the generation of mycotoxins by artificially inoculated colonies with the qualitative composition specific for the strains used.

Maize grain

The maize grain used in the experiment originated from a crop located in Poland, Podkarpackie Voivodeship, Jarosław County (50.00360° N, 22.98822° E). The maize variety was DKC 3972 (FAO 270), sown on approximately April 29, 2024. The following treatments were applied before sowing: urea with a urease inhibitor (200 kg/ha), Omya Calciprill lime (granular chalk, 500 kg/ha), and Korn-Kali (a potassium–magnesium fertilizer containing 40% K₂O as potassium chloride, 6% MgO as kieserite, 4% Na₂O, and

Maize grain contamination

The samples of maize grains were infected with three *Fusarium* fungi, such as *F. avenaceum* (Fa), *F. graminearum* (Fg) and *F. proliferatum* (Fp), according to the procedure described in Section 2.3. After a month all samples aside the control were completely covered with fungi tissue, which mimicked the worst scenario during grain storage under real storage conditions, where all grains are strongly infected.

Fungi and their metabolites

The analytical methods applied in this study allowed for the detection of mycotoxins in

Conclusions

Three different *Fusarium* strains were used to artificially colonize the maize grain. These strains generated a number of mycotoxins with the highest variety of metabolites observed for the *Fusarium proliferatum* strain. Furthermore, in the present study, it was demonstrated that ozonation at relatively low concentrations (150–300 ppm) and short exposure times (15–30 min) was sufficient to achieve complete degradation of a wide spectrum of *Fusarium*-derived mycotoxins in maize. Importantly, no

CRediT authorship contribution statement

Piotr Antos: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Resources, Visualization, Writing – original draft, Writing – review & editing. **Justyna Szulc:** Investigation, Methodology, Resources. **Paweł Wywrocki:** Investigation. **Bożena Kusz:** Methodology, Resources. **Krzysztof Tereszkievicz:** Writing – review & editing. **Radosław Józefczyk:** Investigation, Visualization. **Tomasz Piechowiak:** Formal analysis, Investigation, Methodology, Resources, Visualization,

Declaration of competing interest

Herby authors declare that there are no conflicts of interest to declare.